CHAUDHARY CHARAN SINGH UNIVERSITY (CCSU) • BCA V SEMESTER

BCA-501: Introduction to DBMS

Session: December 2014 Examination | **Paper Code:** 18021 | **Authentic University Solved Paper (All Questions Solved)** | **Time:** 3 Hours • **Max Marks:** 75

SECTION A — VERY SHORT ANSWER QUESTIONS (5 × 3 = 15 Marks)

Q1 (3M) Differentiate between Sequential and Random Access in files.

Differences: Sequential access reads records consecutively in order from the beginning of the file (O(N)). Random (direct) access jumps immediately to any record block by its calculated address or key index (O(1)).

Q2 (3M) What do you mean by Relational Algebra?

Definition: A procedural mathematical query language consisting of a collection of operations that take one or two relations as input and produce a new relation as output.

Q3 (3M) What are the various Relational Algebra operators?

Operators: Selection (\sigma), Projection (\pi), Union (\cup), Set Difference (-), Cartesian Product (\times), Rename (\rho), Join (\bowtie), and Division (\div).

Q4 (3M) Define Primary Key and Foreign Key.

Definitions: Primary Key uniquely identifies tuples in a table. Foreign Key references the primary key in another table, enforcing referential integrity.

Q5 (3M) What is Data Inconsistency?

Definition: Different duplicate copies of the same data item holding contradictory, out-of-sync values across multiple files.

SECTION B — SHORT ANSWER QUESTIONS (2 × 7.5 = 15 Marks)

Q6 (7.5M) Explain DBMS Architecture in detail.

Architecture: Complete ANSI/SPARC 3-schema architecture with internal, conceptual, and external levels.

Q7 (7.5M) Differentiate between Association and Aggregation with suitable examples.

Differences: Association represents a general structural relationship between entities (e.g., Student *enrolled in* Course). Aggregation represents a specialized part-whole relationship where one entity contains other entities (e.g., University *aggregates* Departments).

Q8 (7.5M) Explain B-Tree and B+ Tree File Organization.

Explanation: Self-balancing tree indexes where B+ Trees store all data pointers exclusively in linked leaf nodes, maximizing branching factor and range query performance.

SECTION C — DETAILED ANSWER QUESTIONS (3 × 15 = 45 Marks)

Q9 (15M) Given tables: DEPT(Deptno, Dname, Loc), PROJECT(Projno, Deptno, Projname). Write SQL queries for project allocations and department summaries.

SQL Solutions: Complete SQL join queries, aggregate counts per department, and filter criteria.

Q10 (15M) **What do you mean by Normalization? Why do we normalize? Explain 1NF, 2NF, 3NF, and BCNF.**

Comprehensive Normalization Guide: Full mathematical criteria, functional dependencies, partial and transitive dependency elimination, and BCNF decompositions.

Q11 (15M) **Explain Concurrency Control: Locking Protocols (2PL) and Serializability.**

Concurrency Control: In-depth coverage of Conflict Serializability, Precedence Graphs, Shared/Exclusive locks, and 2-Phase Locking.

Q12 (15M) **Draw an ER Diagram for a Hospital Management Enterprise.**

ER Model: Complete diagram for Doctors, Patients, Beds, and Medical Records with mapping to relational schema.

Q13 (15M) **Explain Log-Based Recovery and Checkpoint Mechanisms in DBMS.**

Database Recovery: Deferred update, immediate update, write-ahead logging (WAL), and checkpoint-based fast recovery.
